

A HYBRID CNN-SNN FRAMEWORK FOR BIOMEDICAL IMAGE CLASSIFICATION USING SADP LEARNING

A Dissertation Submitted
in Partial Fulfilment of the Requirements
for a Minor Degree of

MINOR PROJECT

in **Data Science**

by

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to

**SCHOOL OF DATA SCIENCE
INDIAN INSTITUTE OF SCIENCE EDUCATION AND RESEARCH
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DECLARATION

I, **Athira Chandrasekharan** (Reg. No.: **IMS22070**), hereby declare that this dissertation entitled “**A Hybrid CNN-SNN Framework for Biomedical Image Classification Using SADP Learning**”, submitted to Indian Institute of Science Education and Research, Thiruvananthapuram, towards the partial requirement of **MINOR PROJECT** in **Data Science**, is a bona fide record of original work carried out by me under the supervision of **Dr.Saptarshi Bej**.

This dissertation has never been submitted in part or in full for a degree, diploma or fellowship to this or any other university before. I duly acknowledge all external contributions, statements, datasets or results used, and have listed their sources with adequate detail in the bibliography.

.....
Athira Chandrasekharan

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In my capacity as the project supervisor for the aforementioned candidate, I confirm that the above statements by the candidate are true to the best of my knowledge.



.....
Dr.Saptarshi Bej

Project Supervisor

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ABSTRACT

Student Name: **Athira Chandrasekharan**

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Neuromorphic computing, an emerging field that aims to mimic brain function, is one such field today. It is widely researched for its ability to realize low-power computing through neuromorphic chips. Spiking Neural Networks model the neurons and synapses of the brain in this area. One of the commonly used bio-inspired learning rules in SNNs is Spike-Timing-Dependent Plasticity (STDP). While STDP has its own merits, it lacks the speed and accuracy to make SNNs competitive with respect to conventional AI models such as CNNs. Moreover, such learning rules have not yet been tested in biomedical classification problems. This work evaluates the supervised SADP algorithm on biomedical image datasets across grayscale and color inputs (Lung and colon histopathology and Brain MRI). The goal is to determine whether feature extraction with a CNN improves accuracy compared to simple Poisson encoding. It also examines how the learning rule performs on colored and grayscale images. SADP (Spike Agreement-Dependent Plasticity) is a new learning rule proposed by the lab. It focuses on the agreement between input and output spike trains using Cohen's kappa rather than spike timing [1]. Here, we use a supervised version of this learning rule that uses a CNN to extract features, which, as shown in the results, greatly improves classification accuracy and speed compared to simple Poisson encoding. This work implies that the new learning rule can enable low-power computing to classify biomedical images, such as histopathology or MRI scans, more easily, thereby facilitating diagnosis.

Keywords:

Neuromorphic Computing, Spiking Neural Networks, Spiking Timing Dependent Plasticity (STDP), Spike Agreement Dependent Plasticity (SADP), CNN, Poisson encoding, Hebbian Learning

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Chapter 1

Introduction

Neuromorphic computing tries to mirror how the brain processes information. It builds hardware and software that mimic neurons and synapses. The signals here are not continuous, but spike trains.

To capture this behavior, the functioning of neurons and synapses is modeled using spiking neural networks (SNNs). Information moves as short, discrete events (spikes), quite unlike those in standard networks.

This leads to sparse computation, with only active neurons firing at a given time. The main advantage is the ability to perform low-power computing. This, along with their ability to adapt in real time, could revolutionize computing.

Still, the technology is in its early days. The hardware is not mature yet, and scaling remains difficult. Models are still being refined, and wider adoption will likely take time [?].

In this context, learning in SNNs is often governed by Spike-Timing-Dependent Plasticity (STDP), a widely used local, unsupervised learning rule.

If a presynaptic neuron fires first, the synapse strengthens, a process called long-term potentiation (LTP). The connection becomes more likely to transmit future spikes.

If the order is reversed, the synapse weakens, a process called long-term depression (LTD). The pathway becomes less influential. In this way, causality is encoded directly into the learning process.

The weight update is given by:

$$\Delta w_j = \sum_{f=1}^N \sum_{n=1}^N W(t_i^n - t_j^f) \quad (1.1)$$

The learning window shapes this update:

$$W(x) = \begin{cases} A^+ e^{-x/\tau^+}, & x > 0 \\ -A^- e^{x/\tau^-}, & x < 0 \end{cases} \quad (1.2)$$

For positive timing differences, weights increase, with the effect decaying exponentially while still retaining sensitivity to timing.

For negative differences, weights decrease exponentially, with closely spaced spikes having the strongest influence.

The parameters control behavior. A^+ and A^- set the update strength, while τ^+ and τ^- determine the timing sensitivity.

A shorter time gap exerts a greater influence on the weight update, whereas a larger time gap has minimal effect. This ensures that learning remains both local and stable. [2],[3],[4]

Chapter 2

Existing Works and Research Gap

2.1 Existing Research

2.1.1 Classical Spike-Timing-Dependent Plasticity

Spike-Timing-Dependent Plasticity (STDP) is a popular local learning rule used in computational neuroscience that updates synaptic weights based on the relative timing between pre- and post-synaptic spikes. STDP has been widely applied to unsupervised learning tasks. However, STDP comes with challenges: its reliance on accurate spike timing at the millisecond scale and its high computational cost constrain scalability and applicability to real-time applications..

2.1.2 Teacher-Forcing and Target-Driven Learning in SNNs

This supervised learning rule for SNNs involves teacher-forcing or target-driven approaches. In this method, externally applied currents guide postsynaptic neurons or predefined target signals, shaping their firing patterns to match desired spike trains. This enables learning without gradient calculations. The limitations include the use of externally imposed teaching signals, and the performance of such methods is often sensitive to these signals' strength and duration.

2.1.3 Reward-Based and Reinforcement-Driven Plasticity

Another approach to supervised learning in Spiking Neural Networks combines Spike-Timing-Dependent Plasticity (STDP) with reward-based modulation. In this framework, synaptic changes are first implemented locally via STDP and then adjusted

by a reward signal that depends on how closely the network’s output matches the desired response. This mechanism allows the network to gradually align its activity with target behavior while being biological plausible. However, the effectiveness of this approach depends strongly on how rewards are assigned and propagated, and it can be sensitive to design choices.[5]

2.1.4 Spike-Synchrony-Dependent Plasticity

Spike-Synchrony-Dependent Plasticity (SSDP) is a learning mechanism that updates synaptic weights based on the degree of synchronous firing across neurons. It allows the network to learn more structured and biologically meaningful representations. SSDP operates as a local, post-training adjustment that selectively modifies only a subset of parameters. It acts as a regularization mechanism and can be easily combined with backpropagation, as it does not interfere with the forward computation or gradient flow. It enhances learning without requiring major changes to standard training pipelines.[6]

2.2 Research Gaps

The hardware implementation of STDP remains difficult despite its biological plausibility. Even a small change in spike timing can lead to large changes in weight updates, making learning highly unstable. In comparison, backpropagation-based models perform better on most benchmarks and train faster.

Since STDP requires pairwise interactions between spikes, the complexity scales quadratically with spike counts. For large networks, this becomes expensive.

Several extensions of STDP have also been developed. While some, like reversed or mirror STDP, perform better computationally, they are not fully aligned with neurobiological understanding, making their biological validity uncertain. Some other mechanisms make use of synaptic weight sharing to achieve location invariance. However, weight sharing is nonlocal and is not known to occur in biological networks. More local alternatives would be more biologically plausible.

The main issue we are trying to tackle here is that existing spike-based learning approaches lack effective mechanisms for extracting local spatial features from spatial data. Biologically inspired encoding methods, such as Poisson encoding, capture pixel intensities and missing information about the underlying spatial structures and

patterns in the image. This is a significant limitation for spiking neural networks when working with complex datasets.

Finally, even within edge-based representations, not all features are treated equally. Differentiating between horizontal, vertical, and other edge orientations could improve performance. However, such refinements add further complexity to an already constrained framework.

For these reasons, SNNs have not been widely tested on biomedical datasets. As a result, their practical effectiveness in such domains remains unclear.[7][8]

2.3 Problem Statement

A new learning rule, Spike Agreement Dependent Plasticity (SADP), updates weights based on the agreement between pre- and post-synaptic spike trains. It has outperformed STDP in both accuracy and speed. This work evaluates the supervised version of the SADP algorithm on biomedical image datasets. A hybrid SNN-CNN architecture was used, in which convolutional encoders extract features, which are then converted into Poisson spike trains for SNN learning. The objective is to assess its performance on real-world problems, such as biomedical image classification. [9]

Performance on both grayscale and color images is compared. Although the grayscale images represent primarily intensity information, color images introduce a variety of channels and add more structure to the image. Such differences can have an effect on the interpretation of features by the algorithm. This work will look at the response of SADP to these changes. There are two approaches that are taken into consideration. The first one is simple Poisson encoding. This is a straightforward approach but it fails to consider spatial relationships between pixels. The second method uses CNN based feature extraction. This enables the model to understand the spatial patterns. Local representations like edges and textures are included in the representation. This makes the input to SADP informative.

A direct comparison between these two approaches is done. The goal is to determine whether CNN-based features improve performance. It also examines whether the simpler Poisson encoding is sufficient in certain cases.

The study aims to provide a clear answer to a specific question. Does CNN-based feature extraction improve the performance of supervised SADP on biomedical image data? The results are used to identify where such improvements occur and how significant they are.

Chapter 3

Methodology

3.1 SADP Learning

Spike Agreement Dependent Plasticity (SADP) replaces pairwise spike timing with agreement-based learning using Cohen’s κ . [1]

3.1.1 Spike Encoding

Input features are converted into spike trains using Bernoulli sampling:

$$S_{\text{in}}^{(b)}(t, i) \sim \text{Bernoulli}(x_i^{(b)}) \quad (3.1)$$

for sample b , feature i , and timestep t ,

This produces a binary spike tensor, which is the input to the SNN.

$$S_{\text{in}}^{(b)} \in \{0, 1\}^{T \times N_{\text{in}}} \quad (3.2)$$

3.1.2 CNN Feature Extraction

For complex datasets such as the biomedical images used here, a pretrained CNN extracts features before spike encoding. The CNN consists of convolutional layers, ReLU activations, pooling, and a dense projection layer producing:

$$x^{(b)} \in [0, 1]^{N_{\text{in}}} \quad (3.3)$$

The CNN remains frozen during SNN training. All features are min-max normalized to the range $[0, 1]$. [9]

3.1.3 Neuron Dynamics

All neurons are modeled as leaky integrate-and-fire (LIF) units. For layer l , sample b , and timestep t , the membrane potential evolves as

$$V_l^{(b)}(t) = \lambda V_l^{(b)}(t-1) + I_l^{(b)}(t), \quad (3.4)$$

where $\lambda \in (0, 1)$ is the leak factor.

A neuron emits a spike when its membrane potential exceeds a fixed threshold:

$$s_l^{(b)}(t) = \mathbb{I}(V_l^{(b)}(t) > \theta_l). \quad (3.5)$$

After firing, the membrane potential is reset according to:

$$V_l^{(b)}(t) \leftarrow V_l^{(b)}(t) (1 - s_l^{(b)}(t)). \quad (3.6)$$

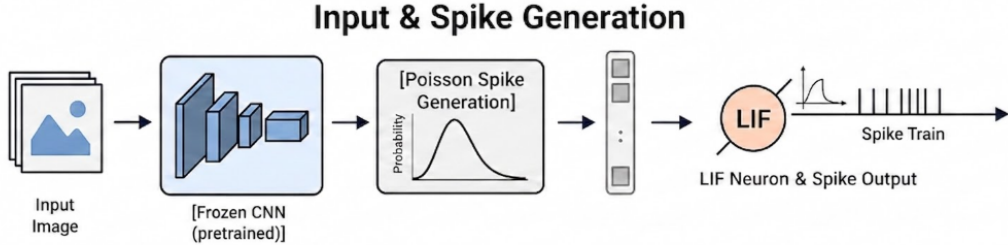


Figure 3.1: Input & Spike Generation pipeline: input image processed by a frozen CNN, converted to Poisson spikes, and passed through an LIF neuron to generate spike trains.

3.2 Network Architectures

3.2.1 1SADP Architecture with One Hidden Layer

The single-hidden-layer architecture consists of weight matrices

$$W_1 \in \mathbb{R}^{N_{\text{in}} \times N_h}, \quad W_2 \in \mathbb{R}^{N_h \times N_{\text{out}}}. \quad (3.7)$$

The input currents are given by

$$I_1^{(b)}(t) = S_{\text{in}}^{(b)}(t)W_1, \quad I_2^{(b)}(t) = s_1^{(b)}(t)W_2. \quad (3.8)$$

We refer to this architecture as **1SADP**.

3.2.2 2SADP Architecture with Two Hidden Layers

The two-hidden-layer architecture introduces an additional weight matrix

$$W_{1,2} \in \mathbb{R}^{N_h \times N_h}. \quad (3.9)$$

The corresponding currents are

$$I_2^{(b)}(t) = s_1^{(b)}(t)W_{1,2}, \quad I_3^{(b)}(t) = s_2^{(b)}(t)W_2. \quad (3.10)$$

No recurrent connections or teacher-forcing currents are used. We refer to this model as **2SADP**.

3.3 Output Decoding

After simulating T timesteps, output spike counts are accumulated as

$$C^{(b)} = \sum_{t=1}^T s_{\text{out}}^{(b)}(t). \quad (3.11)$$

gives the predicted class

$$\hat{y}^{(b)} = \arg \max_k C_k^{(b)}. \quad (3.12)$$

3.4 Learning Rules

The supervision signal is applied only at the output layer and does not propagate across layers, ensuring strict locality of plasticity. All learning rules are local, spike-driven, and gradient-free. Weight updates are applied once per mini-batch.

3.4.1 Supervised Hebbian Learning at the Output Layer

Let $y^{(b)} \in \{0, 1\}^{N_{\text{out}}}$ denote the one-hot class label. The instantaneous error signal is

$$E^{(b)}(t) = y^{(b)} - s_{\text{out}}^{(b)}(t). \quad (3.13)$$

The output-layer weight update is

$$\Delta W_2 = \frac{1}{B} \sum_{b=1}^B \sum_{t=1}^T \left(s_{\text{pre}}^{(b)}(t) \right)^\top E^{(b)}(t), \quad (3.14)$$

where s_{pre} denotes the spike activity of the final hidden layer.

Weights are updated using learning rate η_{out} , multiplicative decay ρ , and clipping:

$$W_2 \leftarrow \text{clip}(\rho W_2 + \eta_{\text{out}} \Delta W_2, [-W_{\text{max}}, W_{\text{max}}]). \quad (3.15)$$

3.4.2 Spike Agreement–Dependent Plasticity (SADP)

Hidden layers are trained using spike agreement–dependent plasticity (SADP), which reinforces neurons whose spike trains agree with the correct-class output neuron beyond chance.

Let

$$s_o^{*(b)}(t) = s_{\text{out}, y^{(b)}}^{(b)}(t) \quad (3.16)$$

denote the spike train of the correct-class output neuron.

For hidden neuron j , the observed agreement is

$$P_{o,j}^{(b)} = \frac{1}{T} \sum_{t=1}^T \mathbb{I}(s_{h,j}^{(b)}(t) = s_o^{*(b)}(t)). \quad (3.17)$$

The marginal firing probabilities are

$$p_{h,j}^{(b)} = \frac{1}{T} \sum_{t=1}^T s_{h,j}^{(b)}(t), \quad p_o^{(b)} = \frac{1}{T} \sum_{t=1}^T s_o^{*(b)}(t), \quad (3.18)$$

and the expected agreement by chance is

$$P_{e,j}^{(b)} = p_{h,j}^{(b)} p_o^{(b)} + (1 - p_{h,j}^{(b)}) (1 - p_o^{(b)}). \quad (3.19)$$

Cohen’s kappa coefficient is computed as

$$\kappa_j^{(b)} = \frac{P_{o,j}^{(b)} - P_{e,j}^{(b)}}{1 - P_{e,j}^{(b)} + \varepsilon}. \quad (3.20)$$

3.4.3 Input-to-Hidden Weight Update

Input-to-hidden synapses are updated using rate-based Hebbian correlations modulated by κ :

$$\Delta W_1 = \frac{1}{B} \sum_{b=1}^B \left(x^{(b)} \right)^\top \kappa^{(b)}. \quad (3.21)$$

Weights are updated with decay and column-wise normalization:

$$W_1 \leftarrow \rho W_1 + \eta_{\text{in}} \Delta W_1, \quad W_{1,:j} \leftarrow \frac{W_{1,:j}}{\|W_{1,:j}\|_2 + \varepsilon}. \quad (3.22)$$

3.4.4 Extension to 2SADP

For the second hidden layer, SADP is applied hierarchically using mean firing rates:

$$\Delta W_{1,2} = \frac{1}{B} \sum_{b=1}^B \left(\bar{s}_1^{(b)} \right)^\top \kappa_2^{(b)}, \quad \bar{s}_1^{(b)} = \frac{1}{T} \sum_{t=1}^T s_1^{(b)}(t). \quad (3.23)$$

Only the output layer receives direct supervision; all preceding layers self-organize through spike agreement. Supervision enters the network exclusively through local mechanisms: (i) error-modulated Hebbian learning at the output layer, and (ii) agreement-based reinforcement using Cohen’s κ in hidden layers.

No backpropagation, surrogate gradients, or teacher forcing are employed.

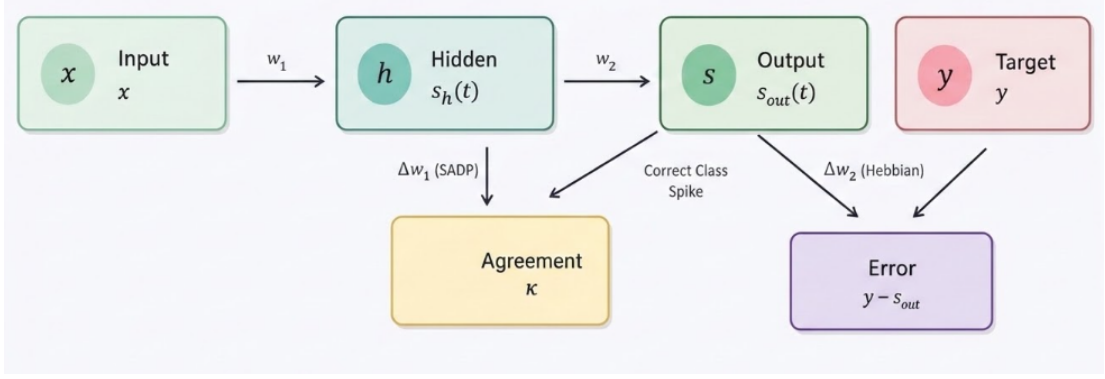


Figure 3.2: Learning framework showing input x , hidden layer activity $s_h(t)$, output spike $s_{out}(t)$, and target y . Weight updates include Δw_1 via SADP (agreement κ) and Δw_2 via Hebbian learning based on output error.

3.5 Evaluation on Biomedical Image Datasets

To evaluate how well the proposed classification framework works, experiments were performed on two biomedical imaging datasets: the LC25000 histopathology dataset [10] and a Brain MRI Tumor dataset [11]. These datasets differ significantly in color, spatial complexity, and diagnostic features, making them a good test of the model’s adaptability. In addition, histopathology and MRI images are commonly used in medical computer vision, which is why they were chosen.

3.5.1 LC25000 Dataset

The LC25000 dataset contains 25,000 color histopathology images, each 768×768 . The images are evenly divided into five classes: lung benign, lung adenocarcinoma, lung squamous cell carcinoma, colon adenocarcinoma, and colon benign tissue. The dataset was first created from 1,250 verified pathology images and then expanded via data augmentation with the Augmentor framework. For this study, the dataset was divided into two subsets:

- **Colon Histopathology:** A binary classification task consisting of colon benign and colon adenocarcinoma images.
- **Lung Histopathology:** A three-class classification task including lung benign tissue, adenocarcinoma, and squamous cell carcinoma.

3.5.2 Brain MRI Tumor Dataset

The Brain MRI Tumor dataset contains grayscale MRI images divided into two classes: tumor and normal. All images were cleaned and prepared using a processing pipeline that removed duplicate images, fixed incorrect labels, and resized them to 224×224 . Data augmentation techniques such as histogram equalization, brightness adjustment, rotation, flipping, and noise addition were also applied. This dataset is a good example of grayscale medical image classification, especially when images have low contrast.

All datasets were evaluated using both Poisson-only encoding and CNN+Poisson encoding schemes. Experiments were conducted across both 1SADP and 2SADP architectures, with temporal resolutions of $T = 25$ and $T = 100$ timesteps.

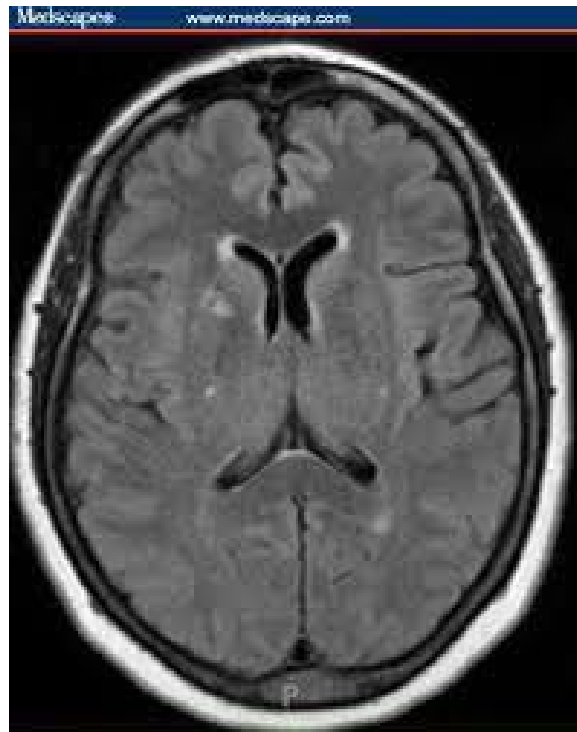


Figure 3.3: Sample Brain MRI image.

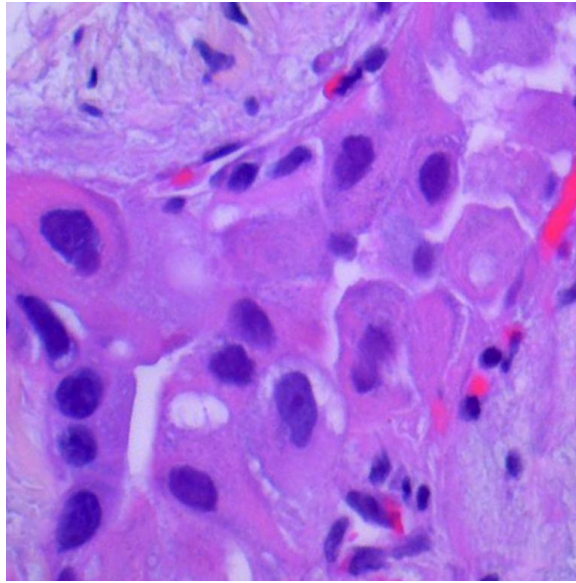


Figure 3.4: Sample Lung histopathology image.

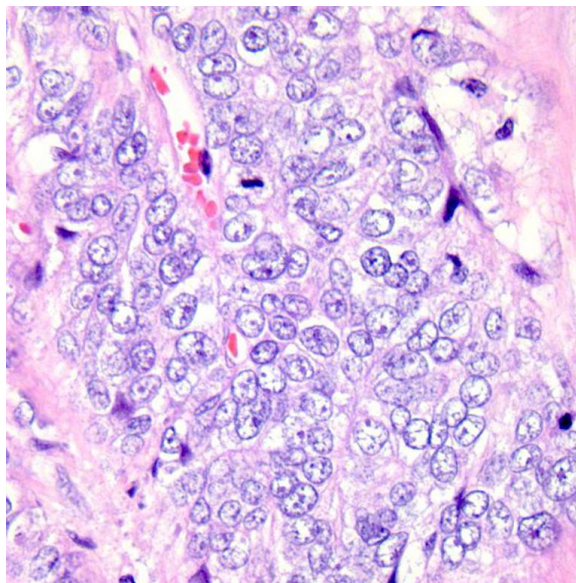


Figure 3.5: Sample Colon histopathology image.

Chapter 4

Results

All datasets were evaluated using:

- Encoding: Poisson and CNN+Poisson
- Architectures: 1SADP and 2SADP
- Timesteps: $T = 25$ and $T = 100$

SADP has linear complexity $O(T)$ per neuron, compared to $O(T^2)$ for STDP, making it more efficient and better suited to neuromorphic hardware.

Results on Biomedical Image Classification

4.1 Overall Performance Trends

Table 4.1 summarizes the performance of SADP-based models across three biomedical imaging datasets. All datasets are evaluated under different architectures, with encodings for chemicals and timesteps. Across all datasets, configurations using **CNN+Poisson encoding** consistently outperform Poisson-only models in both accuracy and macro F1-score. This highlights the importance of structured feature extraction for effective spike-based learning in complex biomedical tasks.

Colon Histopathology For the Colon Histopathology dataset, Poisson-only models perform moderately, with accuracies ranging from 0.63 to 0.66. With CNN-based feature extraction, accuracies significantly improve. The 1SADP model achieves an

Table 4.1: Performance Summary of SADP-based Models on Biomedical Datasets

Dataset	Architecture	Timestep	Encoding	Accuracy	F1-score	Avg Time/Epoch (s)
Colon Histopathology (Color)						
1SADP	25	Poisson-only	0.6600	0.6595	29.34	
1SADP	100	Poisson-only	0.6600	0.6595	45.84	
2SADP	25	Poisson-only	0.6355	0.6232	31.40	
2SADP	100	Poisson-only	0.6355	0.6232	30.36	
1SADP	25	CNN+Poisson	0.9085	0.9079	6.12	
1SADP	100	CNN+Poisson	0.9135	0.9164	3.90	
2SADP	25	CNN+Poisson	0.8690	0.8666	8.43	
2SADP	100	CNN+Poisson	0.9985	0.9985	3.95	
Lung Histopathology (Color)						
1SADP	25	Poisson-only	0.7300	0.7266	60.61	
1SADP	100	Poisson-only	0.7300	0.7266	56.95	
2SADP	25	Poisson-only	0.7380	0.7287	56.67	
2SADP	100	Poisson-only	0.7380	0.7287	43.40	
1SADP	25	CNN+Poisson	0.9650	0.9649	5.99	
1SADP	100	CNN+Poisson	0.9656	0.9637	5.96	
2SADP	25	CNN+Poisson	0.9706	0.9701	5.93	
2SADP	100	CNN+Poisson	0.9843	0.9843	6.09	
Brain Tumor MRI (Grayscale)						
1SADP	25	Poisson-only	0.8586	0.4620	63.27	
1SADP	100	Poisson-only	0.8586	0.4620	71.29	
2SADP	25	Poisson-only	0.8586	0.4620	71.54	
2SADP	100	Poisson-only	0.8586	0.4620	63.53	
1SADP	25	CNN+Poisson	0.9845	0.9688	8.46	
1SADP	100	CNN+Poisson	0.9887	0.9790	8.48	
2SADP	25	CNN+Poisson	0.9827	0.9654	8.44	
2SADP	100	CNN+Poisson	0.9723	0.9487	16.02	

accuracy of 0.9135 at $T = 100$, while the 2SADP model attains near-perfect performance with an accuracy and F1-score of 0.9985. These results suggest that convolutional features effectively capture tissue morphology and color variations that cannot be evaluated through simple spike encoding.

Lung Histopathology A similar pattern is observed in the Lung Histopathology dataset. Poisson-only configurations yield accuracies around 0.73–0.74. CNN+Poisson models achieve accuracies between 0.96 and 0.98, along with great improvements in F1-score. Additionally these improvements are achieved with low computational overhead, as the training time per epoch remains under six seconds for all CNN+Poisson configurations, demonstrating the efficiency of the hybrid CNN-SNN approach.

Brain Tumor MRI In the Brain Tumor MRI dataset, which consists of grayscale images, Poisson-only models achieve stable accuracies around 0.86 but exhibit low F1-scores (0.46), indicating poor class separability. The addition of CNN-based preprocessing significantly enhances both accuracy and F1-score, with CNN+Poisson models reaching accuracies close to 0.99 and F1-scores near 0.98. The results show that spatial feature extraction is crucial even for grayscale medical imaging tasks.

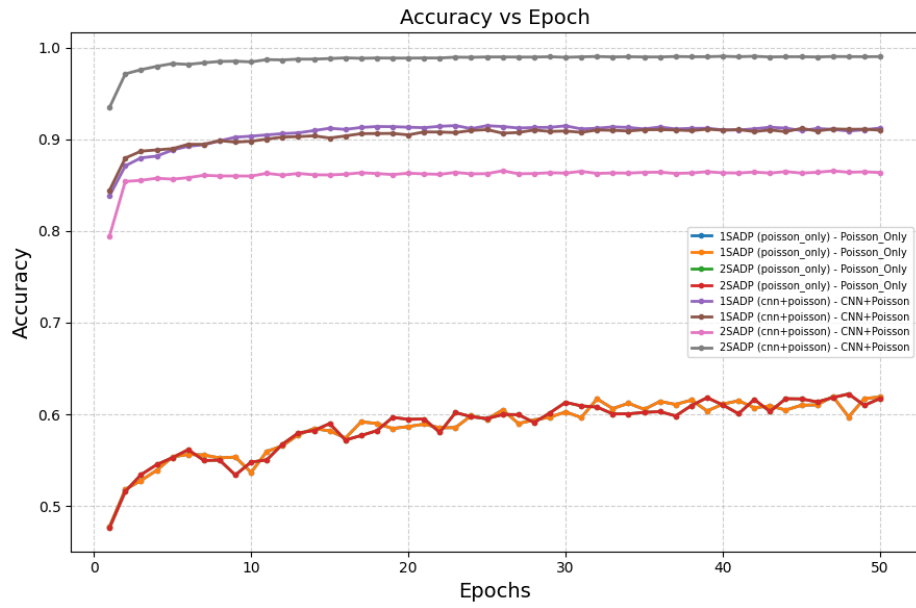


Figure 4.1: Accuracy vs Epoch for the Colon Histopathology dataset.

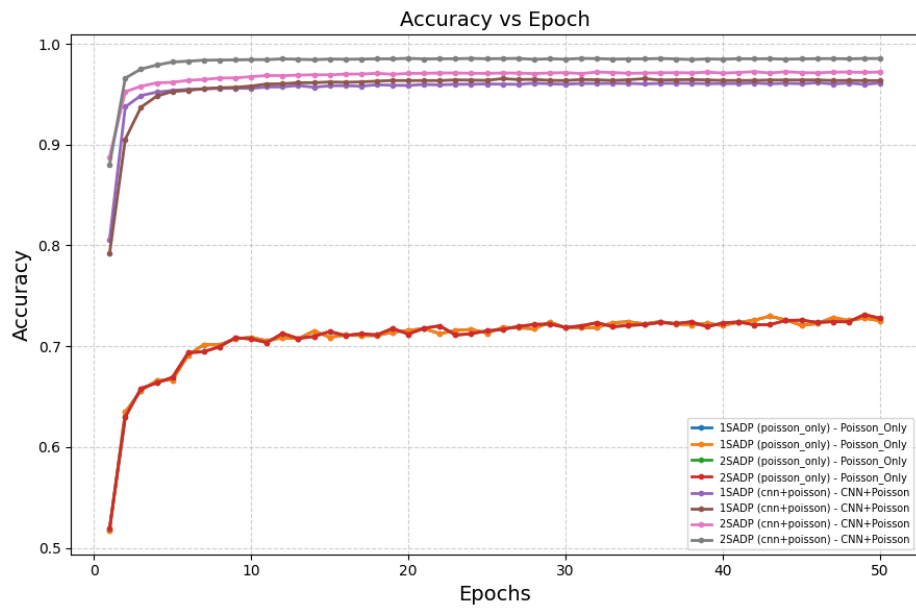


Figure 4.2: Accuracy vs Epoch for the Lung Histopathology dataset.

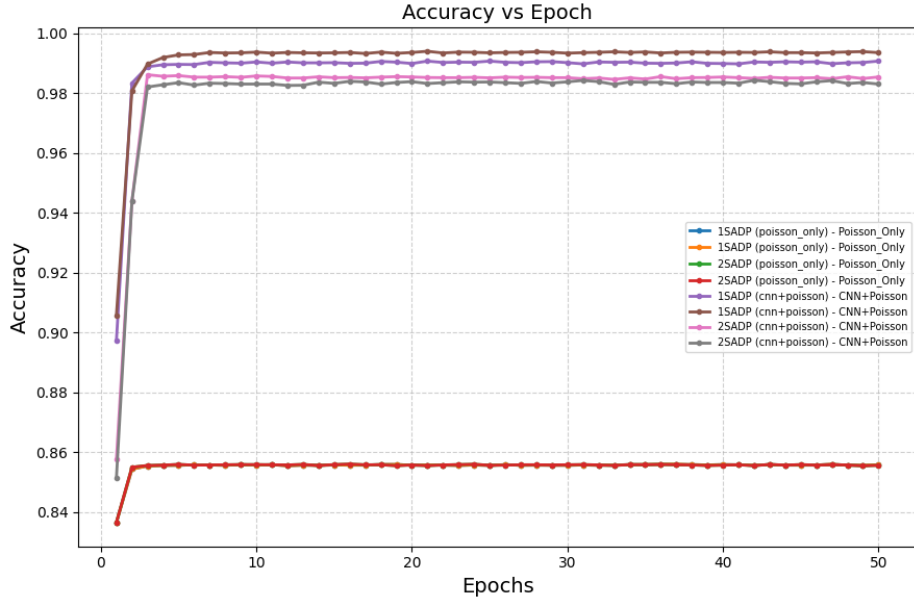


Figure 4.3: Accuracy vs Epoch for the Brain MRI dataset.

4.2 Training Time Comparison

A key objective of this study was to evaluate the computational efficiency of the hybrid CNN-SNN framework compared to traditional Poisson encoding. Figures 4.4, 4.5, and 4.6 illustrate the total training time (in seconds) required for each configuration across 50 epochs.

Across all three datasets, a consistent trend is observed: the **CNN + Poisson** encoding scheme drastically reduces training time compared to the **Poisson Only** method.

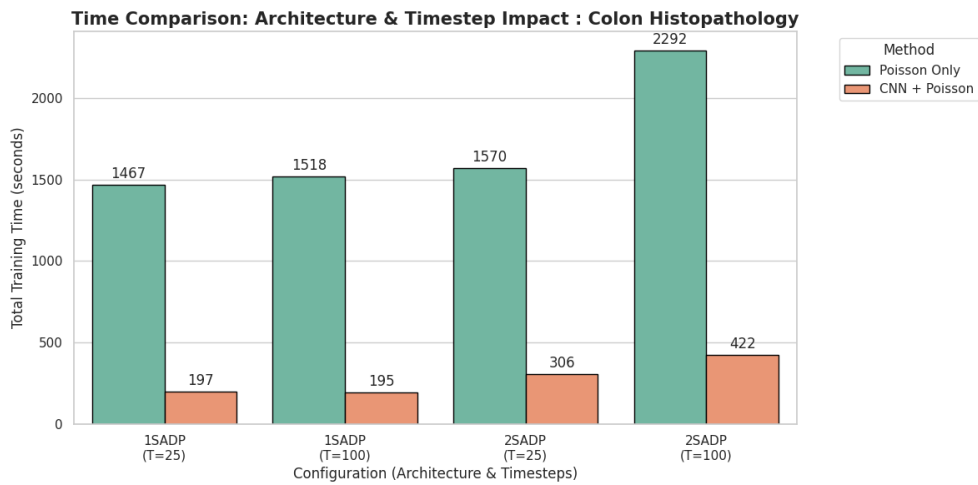


Figure 4.4: Total Training Time Comparison for Colon Histopathology Dataset.

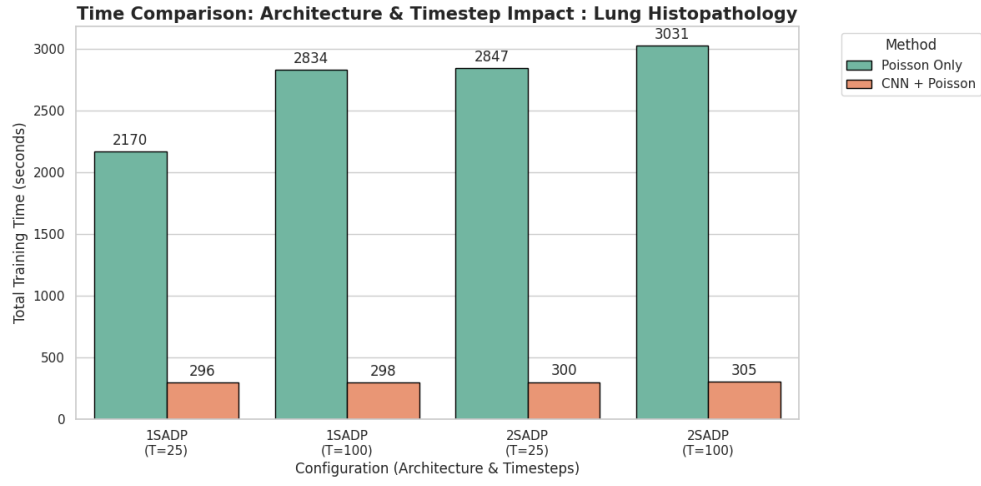


Figure 4.5: Total Training Time Comparison for Lung Histopathology Dataset.

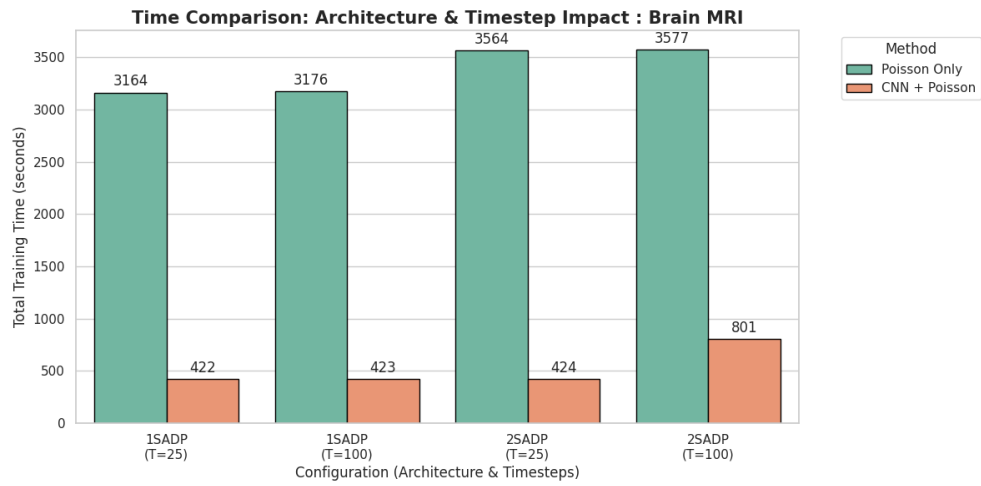


Figure 4.6: Total Training Time Comparison for Brain MRI Dataset.

As shown in the plots:

- **Efficiency Gain:** : In both Colon and Lung data sets, the Poisson only models frequently surpassed the total training time of 3000 seconds. Conversely, the CNN- SNN hybrid models generally took less than 500 seconds to complete the same 50 epochs themselves- a speedup of about 6x to 8x.
- **Architecture Scaling:** The difference between 1SADP and 2SADP architectures in terms of time is marginal compared to the massive difference between the encoding methods. This indicates that the bottleneck in traditional SNN training is the spike-processing of high-dimensional raw input, which the CNN successfully mitigates.

This reduction in time, coupled with the accuracy improvements noted in Table 4.1, proves that the Hybrid CNN-SNN framework is not only more accurate but also significantly more computationally efficient for biomedical image classification.

Preliminary experiments were conducted using supervised STDP on the Colon Histopathology dataset. Supervised STDP showed slow convergence, with each epoch taking approximately 15–30 minutes and training accuracy plateauing near 0.5, indicating limited learning effectiveness.

In addition, attempts were made to implement convolutional spiking neural networks (CSNNs) trained using supervised STDP. These models were found to be computationally prohibitive. Even on simpler datasets such as MNIST, training required several hours (50 hours total runtime without GPU acceleration), making them impractical for larger biomedical datasets in standard simulation environments.

These observations show the limitations of supervised timing-based learning rules in terms of scalability and efficiency.

Chapter 5

Conclusion

5.1 Limitations

Although the proposed hybrid CNN-SNN framework achieves strong performance, it has compromised biological plausibility. Using a pretrained CNN as a feature extractor introduces non-local operations, which are not fully aligned with biological neural systems. In this work, a CNN is used as a fixed encoder to provide structured spatial representations, enabling the study to isolate and evaluate the effectiveness of local spike-based learning rules, such as SADP.

Fully spiking architectures such as convolutional spiking neural networks (CSNNs) offer greater biological realism by learning spatial features directly from spikes. Experiments on CSNN show that these models are computationally intensive and difficult to run in Python-based implementations. It requires significant computational time due to iterative spike-based processing across timesteps, leading to long runtimes for large biomedical datasets.

Hence, the proposed hybrid approach represents a practical compromise between biological inspiration and computational feasibility. Future work includes exploring more efficient spike-based feature extraction methods.

5.2 Discussion

The findings from biomedical data indicate that the supervised SADP model is efficient and flexible for image classification. The performance of CNN-based feature extraction is better when combined with Poisson encoding in colored histopathology

images. This shows that texture and color data play a crucial role in differentiating complex tissue patterns. Conversely, intensity-based encoding fails to capture these details, resulting in lower accuracy.

Applying CNN feature extraction before spike encoding provides a more structured input to the SNN. Neurons that make the greatest contributions to accurate predictions are reinforced, enhancing performance. The near-perfect results on colon and lung histopathology datasets show the advantages of combining CNN features with spike-based learning.

In the Brain Tumor MRI dataset, models with Poisson encoding achieve only moderate accuracy and low class separation. The accuracy can be enhanced by adding CNN-based feature extraction, demonstrating that spatial information plays a significant role even in grayscale images. This further highlights the benefit of well-structured inputs to SNNs.

In general, consistent performance across datasets indicates a high level of generalization. The CNN-SNN hybrid model can extract meaningful features at an early stage, enabling SADP to effectively process complex images without compromising its biologically inspired approach.

In addition to improved accuracy, the hybrid CNN-SNN architecture also exhibits increased computational efficiency. As seen in the analysis of training time, Poisson encoding takes significantly longer. The CNN-SNN hybrid models, in contrast, take approximately 500 seconds to train and achieve a speedup of approximately 6-8x. This implies that the main computational bottleneck in traditional SNN training is the representation of high-dimensional raw input (via spike encoding). Early feature extraction, which is performed by a CNN, reduces dimensionality and complexity of inputs, thus allowing the SNN to learn spikes efficiently. Along with the increase in classification performance, these results confirm that the hybrid CNN-SNN scheme is not only accurate but also more computationally efficient, and hence more suitable to practical biomedical image classification problems.

Practically, it can be implemented to the manufacture of low-power, portable medical equipment. More effective, quicker classification may assist in the simplification of diagnostic instruments, making them more accessible. It is worth noting that this framework combines neuromorphic learning with structured feature extraction without the use of backpropagation. The training process is computationally efficient and easy, and yet has good performance. The approach, in a more general sense, provides a smaller scale analog of conventional deep learning. Neuromorphic hardware can be more easily realized with local learning rules and is more analogous to biological systems.

Such approaches could lead to intelligent diagnostic systems that are not only efficient, adaptable, and applicable in diverse environments but also promising for the future of medical image analysis and decision support.

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